GateMate™ FPGA

PHY Interface for the PCIe Architecture

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DLL Port List

PHY Interface

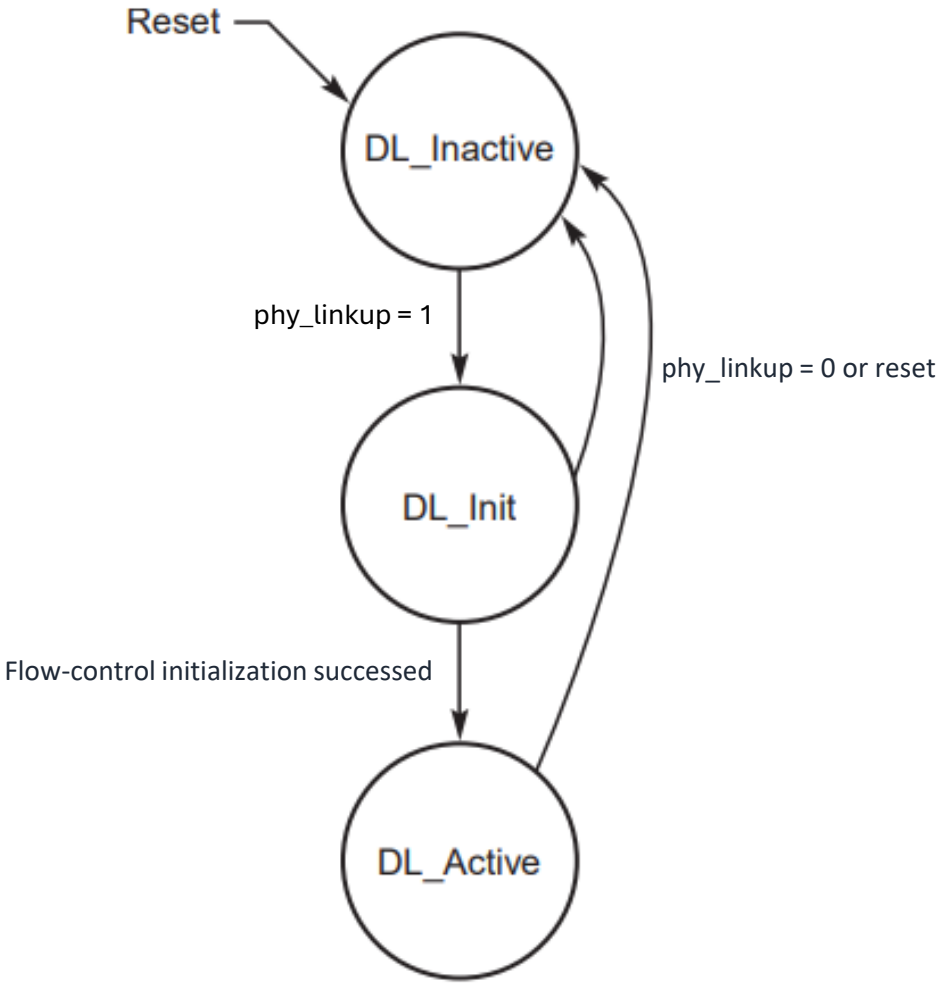
Direction	Name	Description	Width
Input	phy_clk	PHY-domain clock	1 bit
Input	phy_reset	Async reset from the physical layer	1 bit
Input	phy_linkup	Link-up flag indication from the physical layer (LTSSM in L0 state)	1 bit
Input	phy_rx_data	Receive data stream from the physical layer	DATA_WIDTH
Input	phy_rx_data_k	K-character (control) markers for phy_rx_data	DATA_WIDTH/8
Output	phy_tx_data	Transmit data stream to the physical layer	DATA_WIDTH
Output	phy_tx_data_k	K-character (control) markers for phy_tx_data	DATA_WIDTH/8

DLL Port List

TL Interface

Direction	Name	Description	Width
Input	tl_tx_data	TLP payload from TL	DATA_WIDTH
Input Input Input	tl_tx_tlp_valid/ready tl_tx_tlp_first tl_tx_tlp_last	Valid/Ready flag for the TLP payload First batch of the payload from TL Last batch of the payload from TL	1 bit 1 bit 1 bit
Input	tl_tx_hdr_credit	Header credits of DLLP from TL	8 bits
Input	tl_tx_data_credit	Data credits of DLLP from TL	12 bits
Input	tl_tx_update_type	Update type of DLLP from TL	2 bits
Input	tl_tx_packet_type	Packet type of DLLP from TL	2 bits
Input	tl_tx_dllp_valid	DLLP tx valid	1 bit
Output	tl_rx_data	Extracted TLP payload forwarded to TL	DATA_WIDTH
Output Output Output	tl_rx_tlp_valid/ready tl_tx_tlp_first tl_tx_tlp_last	Valid/Ready flag for the received TLP payload First batch of the payload to TL Last batch of the payload to TL	1 bit 1 bit 1 bit
Output	tl_rx_hdr_credit	Extracted header credits forwarded to TL	8 bits
Output	tl_rx_data_credit	Extracted data credits forwarded to TL	12 bits
Output	tl_rx_update_type	Extracted update type forwarded to TL	2 bits
Output	tl_rx_packet_type	Extracted packet type forwarded to TL	2 bits
Output	tl_rx_dllp_valid	DDLP rx valid	1 bit

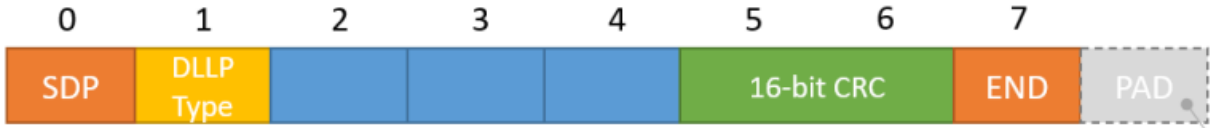
DLL Init FSM


Reset
DL_Inactive
phy_linkup = 1
phy_linkup = 0 or reset
DL_Init
Flow-control initialization successed
DL_Active

Flow-control initialization

- All TLPs are blocked.
- DL_Init has two states
  - DL_FCINIT1:
    - Transmit and Receive three InitFC1 DLLPs:
      - InitFC1-P
      - InitFC1-NP
      - InitFC1-Cpl
    - Repeat the transmission every 34us if both conditions are not met.
    - Record the FC unit values and move onto DL_FCINIT2.
  - DL_FCINIT2:
    - Transmit and Receive three InitFC2 DLLPs:
      - InitFC2-P
      - InitFC1-NP
      - InitFC1-Cpl
    - Repeat the transmission every 34us if both conditions are not met.
    - Ignore the FC unit values and move onto DL_Active

DLLP structure



0000 0000: ACK
0001 0000: NAK
0010 0000: PM_Enter_L1
0010 0001: PM_Enter_L23
0010 0011: PM_Active State_Request_L1
0010 0100: PM_Request_Ack
0011 0000: Vendor specific
0100 0xxx: InitFC1-P (xxx = virtual channel #)
0101 0xxx: InitFC1-NP
0110 0xxx: InitFC1-Cpl
1100 0xxx: InitFC2-P
1101 0xxx: InitFC2-NP
1110 0xxx: InitFC2-Cpl
1000 0xxx: UpdateFC-P
1001 0xxx: UpdateFC-NP
1010 0xxx: UpdateFC-Cpl

$$G(x) = x^{16} + x^{12} + x^3 + x + 1$$

Initial value is FFFFh
DLLP bytes fed in LSb first
Bytes of result inverted and bit reversed

PADs to link width if no new DLLP or TLP to transmit

SDP = 0101 1100
END = 1111 1101

- PM and Vendor specific are not supported in this first version of DLL

DLLP structure

0100b = InitFC1-P
0101b = InitFC1-NP
0110b = InitFC1-Cpl
1100b = InitFC2-P
1101b = InitFC2-NP
1110b = InitFC2-Cpl
1000b = UpdateFC-P
1001b = UpdateFC-NP
1010b = UpdateFC-Cpl

- DLLP Type = {tl_tx/rx_packet_type (2 bits), tl_tx/rx_update_type (2 bits), 1'b0, virtual channel number (3 bits)}
- R are reserved bytes and not considered here
- HdrFC is forwarded to TL via *tl_tx_hdr_credit*
- DataFC is forwarded to TL via *tl_tx_data_credit*

DLLP Handler

- DLLP RX Handler
  - The DLL has a set of registers large enough to store a DLLP. Received data is shifted into these registers.
  - If SDP is detected, extract the DLLP from the register with alignment.
  - The data of DLLP is then extracted and CRC is checked.
  - If CRC check failed, discard the packet.
  - If CRC check passed, forward the data to Transaction Layer with tl_rx_dllp_valid asserted.
  - In case of ACK and NAK.
    - On ACK, release all unacknowledged TLPs at and below of the sequence number stored in ACK. Update ACKD_SEQ to Ack_Seq_Num.
    - On NAK, retransmit all unacknowledged TLPs.
- DLLP TX Handler
  - DLL receives tl_tx_dllp_valid from TL to initiate transmission of a DLLP.
  - Needed data to form a DLLP is also taken from TL. CRC is calculated and added afterward.
  - If LCRC check for a TLP failed, send NAK with the sequence number of the erroneous packet.
  - If LCRC check for a TLP passed, send ACK with the sequence number of the error-free packet.

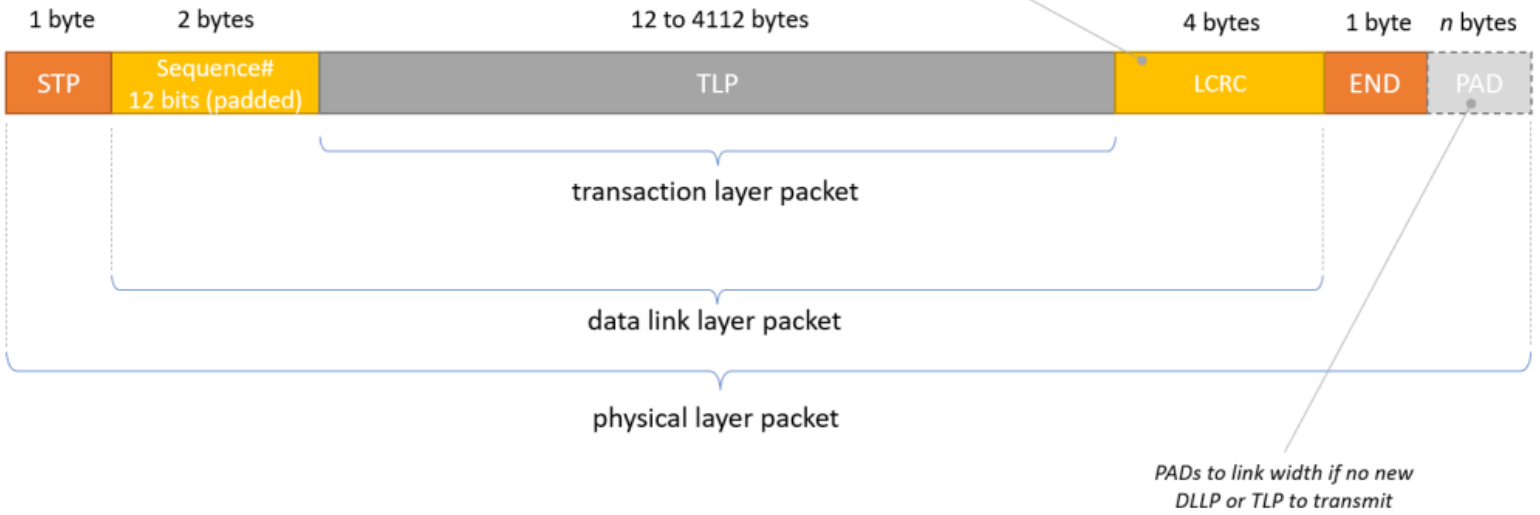
ACK/NAK structure

00000000b = ACK
00010000b = NAK

TLP structure

$$G(x) = x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$$

Initial value is FFFFFFFFh
DLLP bytes fed in LSb first
Bytes of result inverted and bit reversed



STP = 1111 1011
END = 1111 1101
EDB = 1111 1110

TLP Handler

- TLP TX Handler
  - TL indicates the start and end of TLP payload.
  - DLL stores the TLP payload in RAM.
  - Each TLP is assigned to a 12-bit sequence number.
  - TLP is protected by a 32-bit LCRC.
    - The polynomial used is defined in the previous slide.
    - Result of the calculation is complemented and bits of each byte are stored in the reversed order.
  - Priority of transmissions: TLP/DLLP in progress, NAK, ACK, FC DLLP, Retry Buffer re-transmissions, TLP transmission from transaction layer, other DLLP transmission.

TLP Handler

- TLP RX Handler
  - Received TLP payload is stored in RAM, waiting for integrity check.
  - NEXT_RCV_SEQ stores the expected sequence number for the next TLP.
  - NAK_SCHEDULED is asserted when a NAK is scheduled.
    - LCRC not equal, TLP is corrupt, NAK_SCHEDULED is asserted.
    - If TLP sequence number is not equal to the expected value
      - (NEXT_RCV_SEQ - TLP Sequence Number) mod 4096 <= 2048, TLP is a duplicate, an ACK is scheduled.
      - Otherwise, the TLP is out of sequence. NAK_SCHEDULED is asserted.
  - AckNak_LATENCY_TIMER counts time since an Ack or Nak DLLP was scheduled for transmission.
  - If TLP sequence number is equal to the expected value, control data is removed, TLP is forwarded to TL.
    - DLL denotes the start and end of TLP
    - NEXT_RCV_SEQ is incremented
    - If set, the NAK_SCHEDULED flag is cleared
    - ACK is scheduled
  - ACK is sent when: In DL_Active, ACK is scheduled, AckNak_LATENCY_TIMER exceeds value in Table 3-5, NAK_SCHEDULED is cleared.
  - Nullification of TLP with EDB is not supported in this first version of DLL.

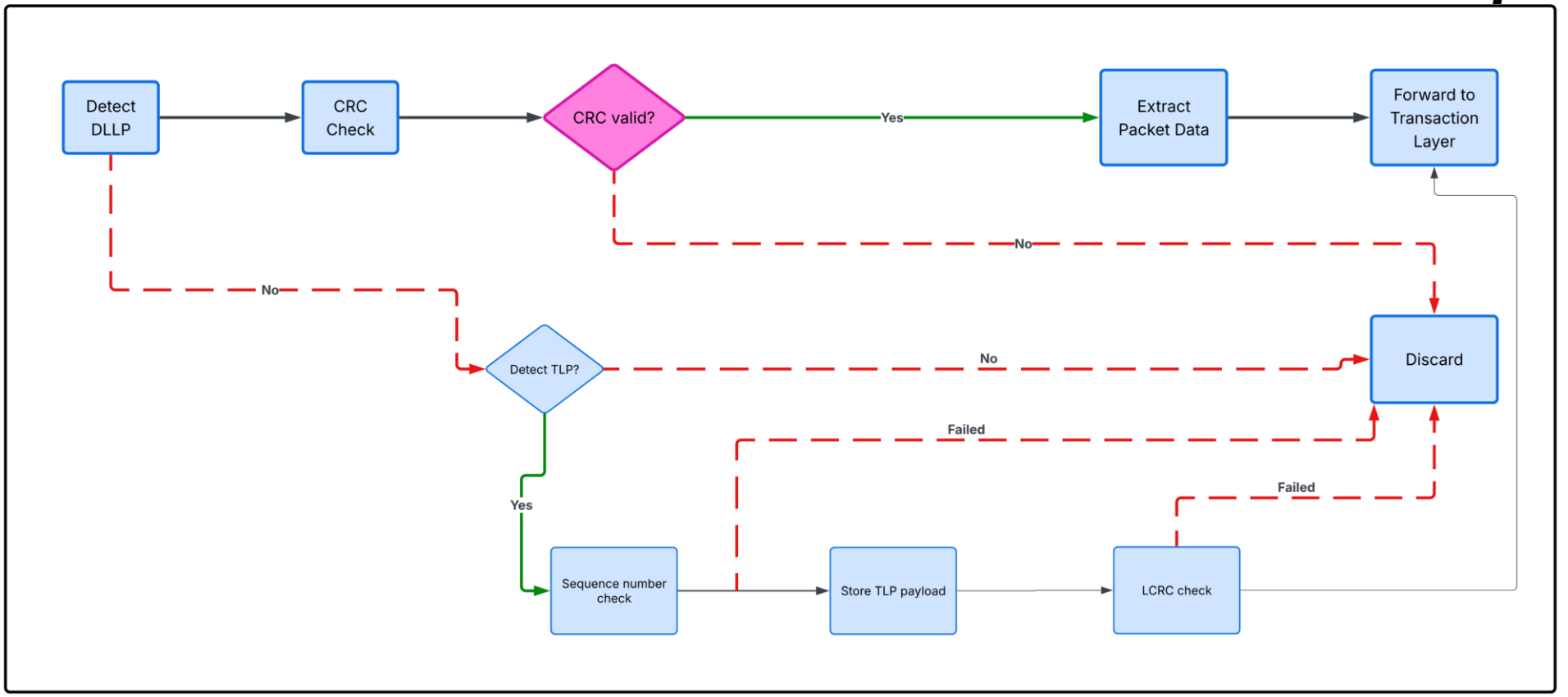
Retry Buffer

- Retry Buffer

- 2-bit replay number counts the number of times the Retry Buffer has been re-transmitted.
  - If replay number roll over from 11b to 00b, signal Physical layer to retrain the link.
- Replay timer counts the time since last transmission or retransmission.
- First version would offer a retry buffer with the size of one.
- Later version might move to a retry buffer with the size of two, implemented with a ping-pong scheme.

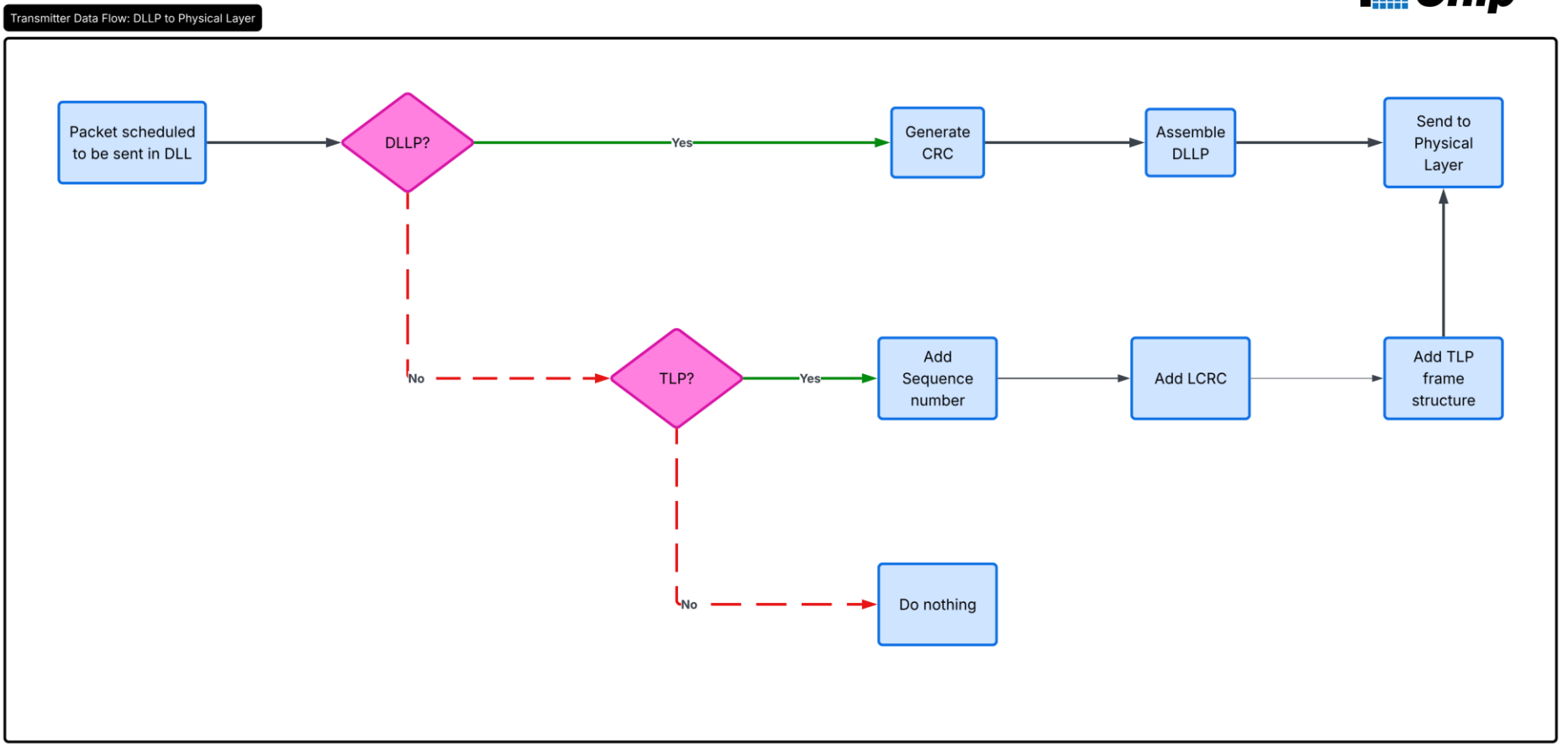
Simple RX flow

Receiver Data Flow: DLLP to Transaction Layer



Also, DLLP Update must be scheduled for sending at regular intervals. This defaults to 30µs (-0%/+50%) but can be configured to be 120µs

Simple TX flow



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